

Laravel migrácie – Informačný systém hudobnej školy

Migrácie sú zoradené podľa závislostí (FK referencie).

2024_01_01_000001_create_users_table.php

php

```
Schema::create('users', function (Blueprint $table) {  
    $table->id();  
    $table->string('name');  
    $table->string('email')->unique();  
    $table->string('password');  
    $table->boolean('must_change_password')->default(true);  
    $table->enum('role', ['admin', 'student', 'teacher', 'guardian']);  
    $table->timestamp('email_verified_at')->nullable();  
    $table->rememberToken();  
    $table->timestamps();  
});
```

2024_01_01_000002_create_departments_table.php

php

```
Schema::create('departments', function (Blueprint $table) {  
    $table->id();  
    $table->string('name'); // hudba, tanec, maľba, ...  
    $table->text('description')->nullable();  
    $table->timestamps();  
});
```

2024_01_01_000003_create_specializations_table.php

php

```
Schema::create('specializations', function (Blueprint $table) {
    $table->id();
    $table->foreignId('department_id')->constrained()->cascadeOnDelete();
    $table->string('name');           // gitara, klavír, spoločenské tance, ...
    $table->timestamps();
});
```

2024_01_01_000004_create_students_table.php

```
php

Schema::create('students', function (Blueprint $table) {
    $table->id();
    $table->foreignId('user_id')->unique()->constrained()->cascadeOnDelete();
    $table->foreignId('specialization_id')->nullable()->constrained()->nullOnDelete();
    $table->string('student_number')->unique();
    $table->date('birth_date');
    $table->timestamps();
});
```

2024_01_01_000005_create_teachers_table.php

```
php

Schema::create('teachers', function (Blueprint $table) {
    $table->id();
    $table->foreignId('user_id')->unique()->constrained()->cascadeOnDelete();
    $table->foreignId('specialization_id')->nullable()->constrained()->nullOnDelete();
    $table->string('employee_number')->unique();
    $table->timestamps();
});
```

2024_01_01_000006_create_guardians_table.php

```
php
```

```
Schema::create('guardians', function (Blueprint $table) {  
    $table->id();  
    $table->foreignId('user_id')->unique()->constrained()->cascadeOnDelete();  
    $table->string('phone')->nullable();  
    $table->timestamps();  
});
```

2024_01_01_000007_create_guardian_student_table.php

```
php  
  
Schema::create('guardian_student', function (Blueprint $table) {  
    $table->foreignId('guardian_id')->constrained()->cascadeOnDelete();  
    $table->foreignId('student_id')->constrained()->cascadeOnDelete();  
    $table->string('relationship')->nullable(); // matka, otec, starý rodič, ...  
    $table->primary(['guardian_id', 'student_id']);  
});
```

2024_01_01_000008_create_classes_table.php

```
php  
  
Schema::create('classes', function (Blueprint $table) {  
    $table->id();  
    $table->string('name'); // napr. "1.A", "Gitara - začiatocní  
    $table->foreignId('teacher_id') // triedny učiteľ  
        ->nullable()  
        ->constrained('teachers')  
        ->nullOnDelete();  
    $table->unsignedSmallInteger('school_year'); // napr. 2024  
    $table->timestamps();  
});
```

2024_01_01_000009_create_class_student_table.php

```
php
```

```
Schema::create('class_student', function (Blueprint $table) {
    $table->foreignId('class_id')->constrained()->cascadeOnDelete();
    $table->foreignId('student_id')->constrained()->cascadeOnDelete();
    $table->primary(['class_id', 'student_id']);
});
```

2024_01_01_000010_create_rooms_table.php

```
php

Schema::create('rooms', function (Blueprint $table) {
    $table->id();
    $table->string('name'); // napr. "Sála A", "Učebňa 101"
    $table->unsignedSmallInteger('capacity')->nullable();
    $table->text('description')->nullable();
    $table->boolean('is_active')->default(true);
    $table->timestamps();
});
```

2024_01_01_000011_create_subjects_table.php

```
php

Schema::create('subjects', function (Blueprint $table) {
    $table->id();
    $table->string('name');
    $table->text('description')->nullable();
    $table->foreignId('department_id')->nullable()->constrained()->nullOnDelete();
    $table->timestamps();
});
```

2024_01_01_000012_create_subject_schedules_table.php

```
php
```

```
Schema::create('subject_schedules', function (Blueprint $table) {
    $table->id();
    $table->foreignId('subject_id')->constrained()->cascadeOnDelete();
    $table->foreignId('class_id')->constrained()->cascadeOnDelete();
    $table->foreignId('teacher_id')->constrained('teachers')->cascadeOnDelete();
    $table->foreignId('room_id')->nullable()->constrained()->nullOnDelete();
    $table->enum('day_of_week', ['mon', 'tue', 'wed', 'thu', 'fri', 'sat', 'sun']);
    $table->time('start_time');
    $table->time('end_time');
    $table->timestamps();

    $table->index(['class_id', 'day_of_week']);
    $table->index(['room_id', 'day_of_week', 'start_time']);
});
```

2024_01_01_000013_create_grades_table.php

```
php

Schema::create('grades', function (Blueprint $table) {
    $table->id();
    $table->foreignId('student_id')->constrained()->cascadeOnDelete();
    $table->foreignId('subject_id')->constrained()->cascadeOnDelete();
    $table->foreignId('teacher_id')->constrained('teachers')->cascadeOnDelete();
    $table->decimal('value', 4, 2); // napr. 1.00 - 5.00 alebo percentá
    $table->string('label')->nullable(); // napr. "A", "1", "Výborný"
    $table->text('note')->nullable();
    $table->timestamp('graded_at')->useCurrent();
    $table->timestamps();

    $table->index(['student_id', 'subject_id']);
});
```

2024_01_01_000014_create_notes_table.php

```
php
```

```
Schema::create('notes', function (Blueprint $table) {
    $table->id();
    $table->foreignId('student_id')->constrained()->cascadeOnDelete();
    $table->foreignId('teacher_id')->constrained('teachers')->cascadeOnDelete();
    $table->text('content');
    $table->enum('type', ['positive', 'negative', 'neutral', 'absence'])
        ->default('neutral');
    $table->boolean('visible_to_guardian')->default(true);
    $table->timestamps();

    $table->index(['student_id', 'type']);
});
```

2024_01_01_000015_create_tasks_table.php

```
php

Schema::create('tasks', function (Blueprint $table) {
    $table->id();
    $table->foreignId('subject_id')->constrained()->cascadeOnDelete();
    $table->foreignId('teacher_id')->constrained('teachers')->cascadeOnDelete();
    $table->string('title');
    $table->text('description')->nullable();
    $table->dateTime('due_at')->nullable();
    $table->enum('type', ['homework', 'test', 'project', 'other'])->default('homework');
    $table->timestamps();

    $table->index(['subject_id', 'due_at']);
});
```

2024_01_01_000016_create_task_submissions_table.php

```
php
```

```
Schema::create('task_submissions', function (Blueprint $table) {
    $table->id();
    $table->foreignId('task_id')->constrained()->cascadeOnDelete();
    $table->foreignId('student_id')->constrained()->cascadeOnDelete();
    $table->decimal('grade', 4, 2)->nullable();
    $table->text('feedback')->nullable();
    $table->timestamp('submitted_at')->nullable();
    $table->timestamps();

    $table->unique(['task_id', 'student_id']); // jeden žiak = jeden odovzdanie na
});
```

2024_01_01_000017_create_instruments_table.php

php

```
Schema::create('instruments', function (Blueprint $table) {
    $table->id();
    $table->string('name'); // napr. "Gitara klasická"
    $table->string('brand')->nullable();
    $table->string('serial_number')->nullable()->unique();
    $table->enum('condition', ['excellent', 'good', 'fair', 'needs_repair'])
        ->default('good');
    $table->foreignId('specialization_id')
        ->nullable()
        ->constrained()
        ->nullOnDelete();
    $table->boolean('is_available')->default(true);
    $table->text('notes')->nullable();
    $table->timestamps();
});
```

2024_01_01_000018_create_room_reservations_table.php

php

```
Schema::create('room_reservations', function (Blueprint $table) {
    $table->id();
    $table->foreignId('room_id')->constrained()->cascadeOnDelete();
    $table->foreignId('reserved_by')->constrained('users')->cascadeOnDelete();
    $table->enum('purpose', ['home_practice', 'club', 'concert', 'lesson', 'other'])
        ->default('lesson');
    $table->dateTime('start_at');
    $table->dateTime('end_at');
    $table->text('note')->nullable();
    $table->enum('status', ['pending', 'approved', 'rejected', 'cancelled'])
        ->default('pending');
    $table->timestamps();

    $table->index(['room_id', 'start_at', 'end_at']);
    $table->index(['reserved_by']);
});
```

2024_01_01_000019_create_instrument_reservations_table.php

php

```
Schema::create('instrument_reservations', function (Blueprint $table) {
    $table->id();
    $table->foreignId('instrument_id')->constrained()->cascadeOnDelete();
    $table->foreignId('reserved_by')->constrained('users')->cascadeOnDelete();
    $table->enum('purpose', ['home', 'club', 'concert'])->default('home');
    $table->dateTime('start_at');
    $table->dateTime('end_at');
    $table->text('note')->nullable();
    $table->enum('status', ['pending', 'approved', 'rejected', 'returned'])
        ->default('pending');
    $table->timestamps();

    $table->index(['instrument_id', 'start_at', 'end_at']);
    $table->index(['reserved_by']);
});
```

2024_01_01_000020_create_events_table.php

php


```
Schema::create('events', function (Blueprint $table) {
    $table->id();
    $table->string('name');
    $table->enum('type', ['concert', 'trip', 'competition', 'exhibition', 'other'])
        ->default('concert');
    $table->dateTime('starts_at');
    $table->dateTime('ends_at')->nullable();
    $table->foreignId('room_id')->nullable()->constrained()->nullOnDelete();
    $table->string('location')->nullable(); // pre výlety/externé udalosti
    $table->text('description')->nullable();
    $table->boolean('is_public')->default(false);
    $table->timestamps();

    $table->index(['type', 'starts_at']);
});
```

2024_01_01_000021_create_event_participants_table.php

php

```
Schema::create('event_participants', function (Blueprint $table) {
    $table->foreignId('event_id')->constrained()->cascadeOnDelete();
    $table->foreignId('user_id')->constrained()->cascadeOnDelete();
    $table->enum('role', ['performer', 'attendee', 'organizer', 'chaperone'])
        ->default('attendee');
    $table->text('note')->nullable();
    $table->primary(['event_id', 'user_id']);
});
```

Poradie spustenia (php artisan migrate)

Migrácie sa spúšťajú v poradí podľa dátumu v názve súboru. Závislosť tabuliek:

users

```
|— departments
|   └─ specializations
|       └─ students — guardian_student
|           └─ class_student
|       └─ teachers
|— guardians — guardian_student
|— classes (→ teachers)
|— rooms
|— subjects (→ departments)
|   └─ subject_schedules (→ classes, teachers, rooms)
|   └─ grades (→ students, teachers)
|   └─ tasks (→ teachers)
|       └─ task_submissions (→ students)
|   └─ notes (→ students, teachers)
|— instruments (→ specializations)
|— room_reservations (→ rooms, users)
|— instrument_reservations (→ instruments, users)
└─ events (→ rooms)
    └─ event_participants (→ users)
```